

Software Fault Prediction using Wrapper based Feature Selection Approach employing Genetic Algorithm

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Abstract—Software fault prediction helps in early identification of software faults and as a result it improves the software quality. It uses previous software metrics and fault data as independent features, to detect whether there is a fault in the software or not. Early prediction of software faults saves a lot of money and effort required to correct those faults. But, as the amount of data is very huge, it is essential for feature selection to get the most useful information. In this paper, we proposed a Genetic Algorithm-based feature selection method that identifies the most useful subset of features for classification purposes. We used a combination of Genetic Algorithm with KNN Classifier, Decision Tree Classifier and Naive Bayes Classifier for our experiments. Our results suggest that, by using Genetic Algorithm for feature selection, our prediction accuracy improved in all the three classifiers for all the datasets and also the number of features were reduced.

Keywords—Software fault prediction, features, feature selection, genetic algorithm, classification, KNN, Naive Bayes, Decision Tree

I. INTRODUCTION

In the past decade, due to the rapid growth of technology, people have greatly focused on software-based systems and as a result the software industry is rapidly evolving. Since software development is done on a large scale, and most of the work is done by humans, faults will inevitably occur. Faults can significantly impact the quality of the software, like it may increase the maintenance cost, and also faults in critical software like those used in healthcare and aerospace can have serious consequences. If the fault is detected after the development process, it will cause an overhead on the developers as they will need to re-design some of the software modules, and it increases the cost and effort. Therefore, Software Fault Prediction (SFP) [1] has become one of the most important part in the process of software development.

SFP helps in identification of software faults at an early stage, thereby improving the software quality. SFP uses previous software metrics and fault data as features (independent), to detect software faults. This saves a lot of money, resources, efforts and man-power required to rectify those faults in the software. But the amount of data has become very huge with the process of industrialization. So, the process of feature selection has become essential to get the most useful information from the data.

Feature Selection (FS) [2] is a method which decreases the number of input variables to our Machine Learning (ML) [3, 4] model by using only the relevant data. This is one of the major tasks for the classification [5, 6] problems. FS helps to

get rid of noise in the data and as a result, it improves the predictive performance of classifiers. It helps us to select only the most useful features for our ML model from our entire data. We have three different methods for FS: Wrapper, Filter and Embedded methods [7].

In this paper, we have presented a wrapper based FS method which is based on Genetic Algorithm (GA) [8] that identifies the most useful feature subset for our classification purposes. In a wrapper based FS method, the process relies on specific ML algorithms which we try to fit on a specific dataset. This method follows an approach of greedy search, which uses the evaluation criteria to evaluate each and every possible combination of features. We used fitness values (i.e. error) as the evaluation criteria. In this method we associated GA with a KNN Classifier [9], Naive Bayes Classifier [10] and Decision Tree Classifier [11] separately.

The remaining part of the paper is organised as follows: Section II presents related studies, Section III introduces a GA based FS method and its implementation, Section IV shows the experimental analysis and computation results, Section V gives the conclusion.

II. RELATED RESEARCH

This section provides information on previous work done over the past few decades on various FS approaches. FS is a method in which the number of irrelevant features are reduced from our data by selecting only the relevant data and removing the noise. It is one of the most important process in classification problems as the presence of a huge number of features in a high-dimensional data impacts the performance of the classifier due to noisy, redundant, and irrelevant features. It helps to improve the prediction accuracy by selecting the most optimal features.

Yoshimasa and Akira [12], in their paper, proposed a new FS method for recognition of characters using GA in which, only those genes whose recognition rate of the training samples exceeds a predetermined threshold value are selected as the parent genes, and the reduction rate of the number of features used for recognition is assumed as the fitness score. In experiments to recognize characters with similar shapes, the proposed method achieved a higher recognition rate and a significant reduction in the number of features compared to Fisher's criterion.

Srividya and Arulmozhi [13] introduced FS classification of skin cancer using GA. Skin cancer is one of the most life-threatening forms of cancer. In their work, they proposed a neural network based on GA for FS that retains the useful

information for classification purposes. They associated GA with KNN classifier. The results suggested that neural classifiers based on GA are very effective and robust in finding the optimal subsets of features.

Marjan and Maziar [14] proposed using GA for FS in video concept recognition. Semantic concept recognition in video is recognized as an important research topic in today's multimedia industry, and classification is the most widely accepted method of concept recognition. In their work, they proposed a new method for reducing the complexity of classification and the time needed for training and classification by selecting only the most useful features. GA was applied to select the features for this purpose. The simulation results showed improved classifier behaviour.

Xiang Chen and Yuxiang Shen [15] conducted research on applying FS to error prediction of software using the multi-objective optimization [16]. SFP can proactively identify potentially bad modules and help the software testers to allocate a larger amount of testing resources to those modules. FS is one of the most efficient ways to mitigate the curse of dimensionality problem [17]. The final results showed that the proposed technique had the advantage of choosing fewer features and getting better performance in most of the projects.

Jinsong and Craig [18] proposed a wrapper-based FS using GA and an external classifier for the analysis of large data sets. Data-sets consists of many redundant, irrelevant and noisy features which degrades the accuracy and increases the complexity. Their paper proposed a novel method to investigate whether irrelevant and noisy features are present in the dataset. Experimental results showed that their method was effective in increasing the accuracy of classification.

Andrey and Achmad [19] proposed a FS method using GA to improve the classification of network intrusion detection systems [20]. The NSL-KDD Cup 99 dataset was used for evaluation and also the dataset was modified by taking a look into the recent network attacks in order to make the dataset more suitable to our current situations. Multiple classifiers were used in the dataset and they found that they obtained the best results using random forest [21] in terms of the training time and classification rate.

In our work, we have proposed a wrapper-based method of FS using GA for SFP. For evaluating our proposed method, we used open-source software defect datasets of NASA. The proposed method is explained in the next section.

III. PROPOSED METHODOLOGY

A. GENETIC ALGORITHM

GA is a heuristic search-based optimization technique inspired by nature which was developed by John Holland in the early 1970s. It can be used to find the best solution to a problem. It is based on the mechanism of natural genetics and evolution. It is an evolutionary technique that starts with an initial solution to lookup for an optimal solution till the stopping criteria is satisfied, followed by selection, crossover, and mutation operations. Firstly, a population of N chromosomes $P = [H_1, H_2, H_3, \dots, H_N]^T$ are generated randomly where we can define each chromosome by $H_i = [h_{i,1}, h_{i,2}, h_{i,3}, \dots, h_{i,D}]$. Here, the dimension of each chromosome is represented by D and the number of chromosomes in the population is represented by

$i = 1, 2, 3, \dots, N$. In binary GA, the genes present in the chromosomes are represented by the values 0 and 1. The different steps involved in GA are shown in Fig. 1.

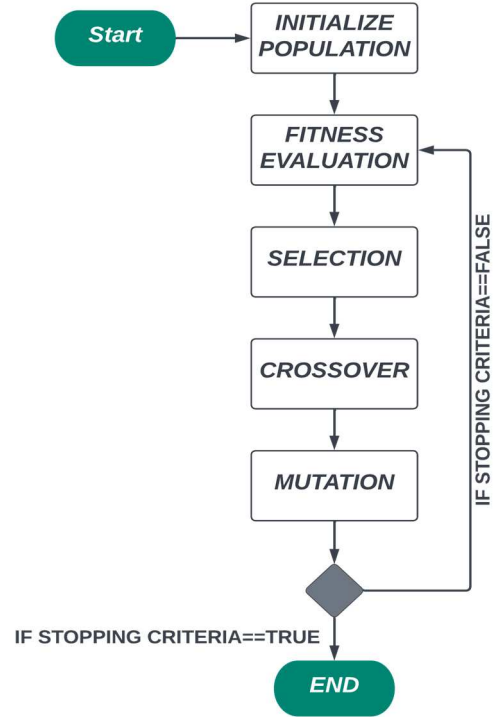


Fig. 1. Different steps in GA

(a) INITIALIZATION PHASE

In the initialization phase, the population which is a subset of all the possible solutions of the problem is built and initialized. We randomly generate the initial population, which is a binary 2-D matrix where, each row and column represents a chromosome and a gene respectively. Here, the total number of columns (genes) equals the number of independent features present in the dataset, where '0' means that the particular feature will not be used by the ML model and '1' means that the particular feature will be used by the ML model. A sample population is shown below in Fig. 2:

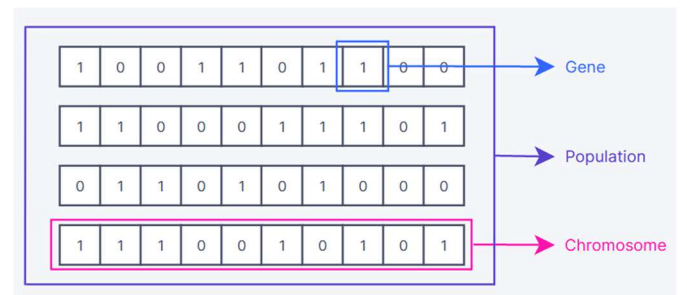


Fig. 2. Sample population

(b) FITNESS CALCULATION

After the population is initialized, the fitness score of each and every chromosome present in the population is calculated. For calculating the fitness, we train our ML model using the training data and only those features, which are represented by each chromosome. The prediction error (fitness value) for each chromosome is stored in an array whose length equals the number of chromosomes present in our population. Only

those chromosomes with low fitness value (error) have a high probability to be selected for recombination.

(c) SELECTION PHASE

The parents are selected on the basis of the fitness score of the chromosomes. On the basis of these fitness scores, a pair of chromosomes with the lowest fitness score is selected. Roulette wheel selection was used for the parent selection. In this method, the probability of a chromosome (individual) being chosen is directly dependent on the fitness score of the individual. Here, a fitter individual will have a greater area on the wheel and therefore a greater chance of being selected.

(d) CROSSOVER

Crossover is done to create children or offspring from one generation to another generation using the selected parents. For our purpose, we used single point crossover. In single-point crossover, we select a point and the parts shared at that point are exchanged between the parent chromosomes to generate the offspring. Then, these new children are added to the population. Single point crossover is shown below in Fig. 3:

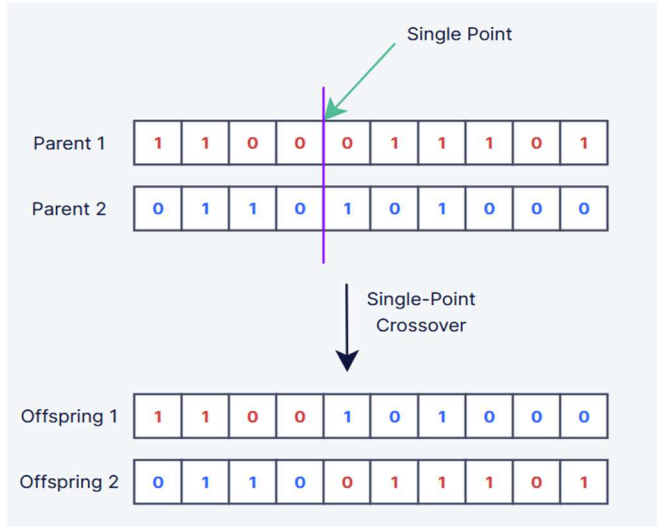


Fig. 3. Single point crossover

(e) MUTATION

Mutation is performed after crossover to bring diversity in the population. We used the Bit Flip method of mutation, in which one or more than one bits are selected and flipped. If the bit which is selected is '0', then it is flipped to '1' and vice-versa. An illustration of mutation is shown below in Fig. 4:

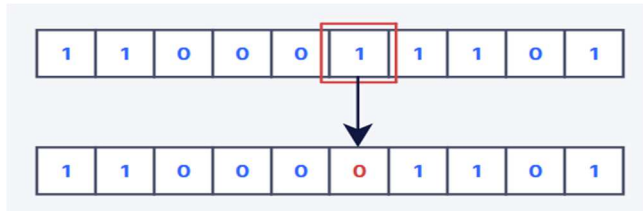


Fig. 4. Mutation

B. USE OF GENETIC ALGORITHM FOR FEATURE SELECTION

FS is a method of identifying the most important features for a ML model. It helps to remove unwanted features, which may reduce the performance of the ML model. FS enhances

the generality performance of a ML model. So, the core idea is to reduce the prediction error to a minimum value. When number of features is very large, the process of FS becomes very difficult. So, some intelligent methods are needed to perform FS.

One of the most useful algorithms for FS is GA. Feature Selection using Genetic Algorithm (FSGA) is an evolutionary method which starts with an initial solution. This method searches for the subset of optimal features until the stopping criterion is met, followed by selection, crossover and mutation operations. Firstly, a population of N chromosomes $P = [H_1, H_2, H_3, \dots, H_N]^T$ are generated randomly in which we can define each chromosome by $H_i = [h_{i,1}, h_{i,2}, h_{i,3}, \dots, h_{i,D}]$. Here, the dimension of each chromosome is represented by D and the number of chromosomes in the population is represented by $i = 1, 2, 3, \dots, N$. In binary GA, the genes present in the chromosomes are represented by the values 0 and 1. The fitness of each chromosome, can be evaluated by using classification algorithms based on selected attributes and can be expressed as $f(H_i)$ by using (1), where $i = 1, 2, 3, \dots, N$.

$$f(H_i) = \frac{\sum E}{P} \quad (1)$$

Here, E is the no. of errors and P is the total number of predictions.

The roulette wheel selection method is used to select the chromosomes based on the probability of each chromosome. The probability of the i^{th} chromosome is computed and expressed as $P(H_i)$ by using (2).

$$P(H_i) = \frac{f(H_i)}{\sum_{i=1}^N f(H_i)} \quad (2)$$

Then, we calculate the cumulative probability and two parents (chromosomes) are selected for crossover based on random number generation between 0 and 1. If the cumulative probability of the i^{th} chromosome is greater than the random number, then that chromosome is selected as the parent for crossover phase.

In the crossover phase, two fit chromosomes are selected and they are combined to generate new children (chromosomes). In this process, a random integer is chosen in the range of $[0 - D-1]$ as crossover point, where each chromosome exchanges its genes with each other.

In the mutation phase, genes are randomly shuffled across chromosomes to reduce the chance of them getting stuck in local optima. The fitness of chromosomes has been computed by considering fitness of N chromosomes to determine next iteration. This process is carried out for a certain amount of iterations and each iteration generates a set of new chromosomes by using crossover, mutation, and evaluation process. The chromosomes having the fastest adoption capability will choose the best chromosome for reproduction in the next cycle.

Our work focuses on how to apply GA to optimize the performance of KNN Classifier, Decision Tree Classifier, and Naïve Bayes Classifier by choosing the most appropriate and useful features. The algorithm for the FSGA approach has been explained in Algorithm 1.

C. TRAINING THE MACHINE LEARNING MODEL AND FITNESS CALCULATION

In every iteration of our FSGA algorithm, the ML model gets trained using the selected subset of features from the training data. Using the trained ML model we predict the output from the testing data. The fitness score (error) is then calculated by comparing the predicted output with the testing class label (which is the actual output of the testing data). Equation (1), which is described earlier, is the equation used for the calculation of fitness.

Algorithm 1: Implementation of FS using GA

1. Initialize the crossover rate (CR), the mutation rate (MR), population size (N) and dimension of the problem (D).
2. Computation of training and testing data.
3. Initialize the initial population randomly with N chromosomes as $P = [H_1, H_2, H_3, \dots, H_N]^T$ and each of the chromosomes is defined as $H_i = [h_{i,1}, h_{i,2}, h_{i,3}, \dots, h_{i,D}]$.
4. Evaluate the fitness (error) for each individual (chromosome).
5. $t=1$.
6. while ($t \leq \text{max_iter}$) do
7. Convert error to accuracy which is the inverse of fitness.
8. Compute the probability of each chromosome.
9. for $i=1$ to N do
10. if $\text{rand}() < CR$
11. Compute cumulative probability (C).
12. Select two parents using Roulette Wheel.
13. Choose one crossover point within the range $[0 - D-1]$.
14. Apply the single-point crossover between two parents.
15. end if
16. end for
17. Apply the mutation (1 to 0 or 0 to 1) based on MR.
18. Compute the fitness of the updated chromosomes.
19. Select N best chromosomes and its fitness.
20. $t=t+1$
21. end while.

IV. RESULT ANALYSIS

A. DATASETS USED

For our experiments we used 12 open-source software defect datasets from the PROMISE repository [22] of NASA:

KC1, KC3, JM1, CM1, MC2, MC1, PC1, PC2, PC3, MW1, PC4, and PC5. The information in details regarding these datasets is given in Table 1.

Table 1. Information regarding the datasets

Sl. No	Data-set	No. of Instances	No. of Features
1	KC1	1183	22
2	KC3	194	40
3	JM1	7782	22
4	CM1	327	38
5	MC1	1988	39
6	MC2	125	40
7	PC1	705	38
8	PC2	745	37
9	PC3	1077	38
10	PC4	1287	38
11	PC5	1711	39
12	MW1	253	38

B. EXPERIMENTAL CONDITION

The experiments were performed in Jupyter Notebook 6.4.8 using Python version 3.9.12. The system hardware configurations were Intel Core i7-10750H processor, 32 GB RAM and NVidia RTX 2070 Max-q GPU. In the experiments, we set the cross-over rate (CR) to 0.8 and the mutation rate (MR) to 0.01. The population-size (NP) was set to 10 and we set the maximum number of iterations for each FS method to 200.

C. EXPERIMENTAL ANALYSIS

Experiments were performed on all the above 12 datasets shown in Table 1. The average fitness (error) plots for all the datasets after performing the experiments using KNN, Decision Tree, and Naïve Bayes Classifier are displayed in Fig. 5. In the graphs, the X and the Y axes represent the number of iterations and the fitness scores respectively and FSKNN, FSDT and FSNB represent the fitness plots for FS using KNN, Decision Tree and Naïve Bayes Classifier respectively.

Table 2 presents the classification accuracy of the classification models and the optimal number of features of the FS using GA approach for all 12 datasets. The values in bold in Table 2 represent the highest classification accuracy using FSGA among the three different classifiers and the optimal number of features selected.

The experimental results of the FSGA model shows that prediction accuracy increases for all the datasets in all three classifiers on using FS. Number of selected features also decreases as it selects only the most useful features and discards the unnecessary features and noise.

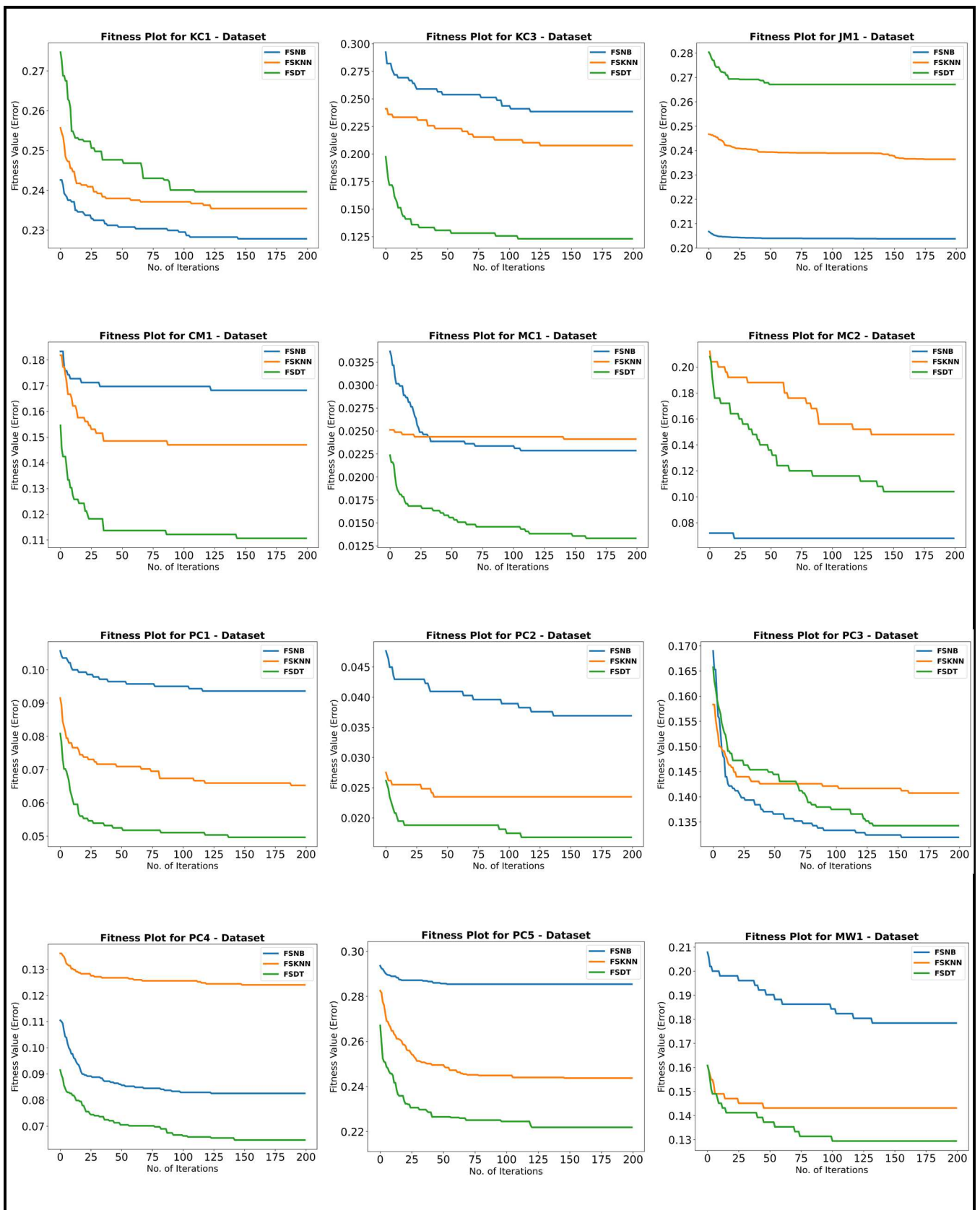


Fig. 5. Fitness (Error) plots for the KC1, KC3, JM1, CM1, MC2, MC1, PC1, PC2, PC3, MW1, PC4, and PC5 datasets

Table 2. The classification accuracy (%) and the number of selected features

Sl. No	Data-set	Classifiers Used	Accuracy (%) without FS	Accuracy (%) with FSGA	No. of selected features
1	KC1	KNN	69.6203	76.4557	8.6
		Decision Tree	72.1519	76.0338	11.6
		Naïve Bayes	74.2616	77.2152	8.4
2	KC3	KNN	74.3589	79.2308	19.4
		Decision Tree	76.9231	87.6923	21.1
		Naïve Bayes	66.6667	76.1538	15.5
3	JM1	KNN	73.3462	76.3584	11.9
		Decision Tree	69.9422	73.2884	11.7
		Naïve Bayes	78.9981	79.6211	10.0
4	CM1	KNN	75.7576	85.3030	18.4
		Decision Tree	80.3030	88.9394	17.7
		Naïve Bayes	77.2727	83.1818	15.3
5	MC1	KNN	96.4824	97.5879	18.4
		Decision Tree	97.7387	98.6683	20.8
		Naïve Bayes	95.7286	97.7136	16.7
6	MC2	KNN	76.0	85.2	16.8
		Decision Tree	68.0	89.6	19.3
		Naïve Bayes	92.0	93.2	18.9
7	PC1	KNN	89.3617	93.4752	17.7
		Decision Tree	88.6525	95.0355	20.0
		Naïve Bayes	87.2340	90.6383	17.2
8	PC2	KNN	96.6443	97.6510	18.2
		Decision Tree	95.3020	98.3221	19.4
		Naïve Bayes	93.9597	96.3087	12.8
9	PC3	KNN	82.4074	85.9259	18.9
		Decision Tree	78.7037	86.5740	19.9
		Naïve Bayes	68.9815	86.8056	13.5
10	PC4	KNN	84.4961	87.5969	17.2
		Decision Tree	91.0853	93.5271	20.0
		Naïve Bayes	86.8217	91.7442	19.2
11	PC5	KNN	67.0554	75.6268	18.3
		Decision Tree	72.5948	77.8134	19.6
		Naïve Bayes	70.5539	71.4577	20.0
12	MW1	KNN	78.4314	85.6863	19.3
		Decision Tree	74.5098	87.0588	17.5
		Naïve Bayes	76.4706	82.1569	18.3

V. CONCLUSION

In this paper, we proposed a FSGA model for SFP datasets. SFP is necessary as it can reduce the costs and efforts required to correct the fault in the software. We used 12 different fault prediction datasets and for each dataset three different classifiers were used. In this paper, the objective is to use GA to effectively find the best and most suitable features. The results obtained from our experiments show that the proposed FSGA model increases the prediction accuracy for all the three classifiers in all datasets. It also ignores the unnecessary features and noise and selects the best suitable and optimal

feature subset. This wrapper-based FS model can be applied to various applications such as disease prediction, character recognition and weather forecasting.

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